

Andre Casagrande Rosenberg

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Profile

Master in Artificial Intelligence student at Atlantis University with academic experience in machine learning, computer vision, Python, PyTorch, Jupyter, and transfer learning. Civil engineer with professional experience in aerial imaging, photogrammetry, GIS, geospatial analysis, mapping, and engineering project delivery.

Education

Master in Artificial Intelligence — Atlantis University · In progress, 2025–2027

Coursework: Machine Learning; Advanced Applied Machine Learning; Natural Language Processing; AI for Human–Computer Interaction; Ethics in AI; Artificial Intelligence.

Bachelor of Civil Engineering — Universidade Vila Velha · Completed August 2020

Selected AI Projects

Concrete Crack Image Classification — Built a MobileNetV2 transfer-learning workflow for binary classification of concrete surface images using a 40,000-image dataset with an 80/20 train/test split.

Local AI Skills Showcase — Built an offline HTML/CSS/JavaScript page that displays six coursework skills from a separate data file and works by opening index.html directly.

Self-Attention in PyTorch — Implemented Query, Key, and Value matrix computation and scaled dot-product self-attention and validated it on a three-token example.

Professional Experience

Owner & Responsible Civil Engineer — BR Solos Engenharia Ltda. · June 2023–Present

Lead engineering and aerial-mapping workflows from field capture through processing, spatial analysis, cartographic outputs, and technical delivery.

Civil Engineer — Autovia Engenharia Ltda. · Aug 2020–May 2021; Dec 2022–Aug 2023

Acquired and processed aerial imagery and field data and produced maps and technical charts for engineering projects.

Partner & Responsible Civil Engineer — T&R Engenharia Ltda. · Apr 2021–Jan 2023

Led engineering responsibility and UAV/aerial data-acquisition activities for mapping, cartography, and project development.

Technical Skills

Python · Jupyter · PyTorch · torchvision · Computer Vision · MobileNetV2 · Machine Learning · UAV / Aerial Imaging · Photogrammetry · GIS / Geoprocessing · Cartography · Spatial Databases · Civil Engineering

Languages

Portuguese · English